

TrES-3b

R-2830

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210814 · created 2026-09-05T23:30:47+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459440.6966 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.021 +/- 0.057
Transit depth (Rp/Rs)^2	0.045 +/- 0.24 [%]
Orbital Inclination (inc)	80.42 +/- 2.9
Airmass coefficient 1 (a1)	1.58 +/- 0.77
Airmass coefficient 2 (a2)	-0.4 +/- 1.12
Scatter in the residuals of the lightcurve fit is	50.62 %
Best Comparison Star	None
Optimal Aperture	2.41
Optimal Annulus	3.0
Transit Duration (day)	0.024 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.021 ± 0.057 vs 0.16309 (deviation 2.49σ)
Duration fit vs published	0.024 d vs 0.0567 d (deviation 1.26σ)
Mid-time O–C	-3.6 min
Depth SNR	0.4
Residual scatter	50.62 %

Light curve

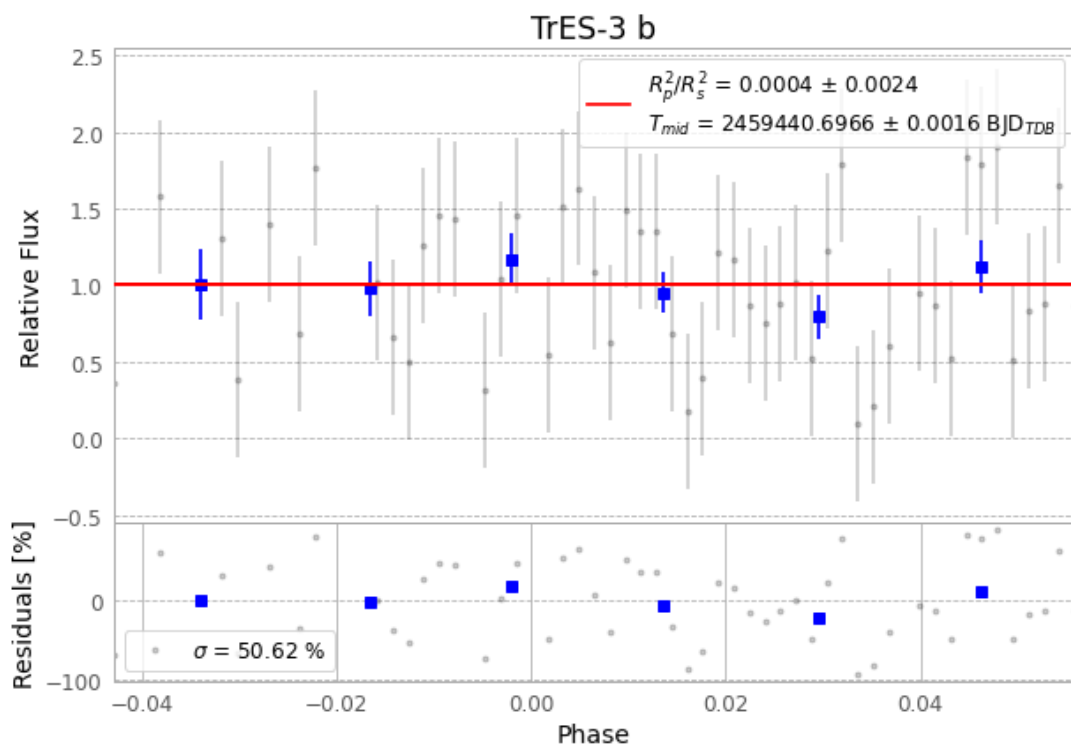


Figure 1 — FinalLightCurve_TrES-3 b_2021-08-13.png (SHA-256 dd1a53bf437dcd44...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [581, 111], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b6321a5a289e4447...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), TRES-3210814031512.FITS → TRES-3210814062412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210814.log (301c388eaa73...), FinalLightCurve_TrES-3 b_2021-08-13.png (dd1a53bf437d...), FinalLightCurve_TrES-3 b_2021-08-13.csv (8c954d1aa7b3...)

Compute resources

Wall time 187.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 23:30Z.